

BANKURA, IndiaWMO: **427060**Lat: **23.383N**Lon: **87.083E**Elev: **100**StdP: **100.13**Time Zone: **5.50 (IND)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.0	12.0	4.3	5.2	25.7	5.8	5.8	22.8	2.4	19.2	2.3	20.2	0.2	N/A	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	13.2	41.4	24.1	39.6	24.9	37.9	25.1	29.1	34.6	28.7	33.8	28.4	33.0	1.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.3	31.4	27.6	23.8	31.0	27.2	23.3	30.5	96.4	35.2	94.5	34.0	92.7	32.9	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	2.4	2.1	1.6	9.0	43.8	0.7	1.8	8.4	45.1	8.0	46.1	7.6	47.1	7.0	48.4	
(5)				7.4	29.9	0.8	0.6	6.8	30.3	6.4	30.6	5.9	30.9	5.4	31.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	19.3	23.2	28.3	31.8	31.5	31.2	29.1	29.1	29.0	27.4	24.0	20.2
	DBStd	4.62	1.93	2.57	2.50	2.71	2.35	2.35	1.29	1.22	1.41	1.79	1.88	2.10
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	21	13	1	0	0	0	0	0	0	0	0	0	7
	CDD10.0	6209	288	371	566	653	666	635	593	593	569	539	420	315
	CDD18.3	3187	42	138	308	403	408	385	335	334	319	281	170	64
	CDH23.3	36813	267	1159	3516	5559	5582	5166	3946	3908	3636	2663	1078	331
	CDH26.7	17709	33	452	1931	3434	3284	2837	1541	1505	1379	963	309	42
Wind	WSAvg	0.4	0.3	0.3	0.5	0.7	0.7	0.6	0.5	0.4	0.3	0.2	0.2	0.3
Precipitation	PrecAvg	1447	20	26	17	49	116	205	371	310	234	74	17	8
	PrecMax	2052	74	75	87	120	246	615	723	495	380	405	130	45
	PrecMin	876	0	2	0	1	45	10	153	64	93	0	1	0
	PrecStd	356	23	20	24	35	56	141	154	106	80	97	28	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.3	35.4	39.8	44.1	42.7	41.8	35.5	34.7	35.2	34.6	32.4	29.5
		MCWB	18.6	20.0	21.6	23.7	24.0	27.3	27.8	28.2	27.6	25.4	22.4	20.2
	2%	DB	27.6	33.1	37.8	42.1	40.2	39.6	33.9	33.6	33.9	33.2	30.8	27.8
		MCWB	17.3	19.6	21.5	23.0	25.7	27.6	28.1	28.1	27.7	25.9	22.3	18.8
	5%	DB	26.1	31.1	36.2	40.3	38.4	37.6	32.9	32.7	32.9	32.1	29.7	26.5
		MCWB	16.9	18.9	21.1	23.3	26.6	27.9	27.9	27.9	27.6	25.7	21.8	18.3
	10%	DB	24.8	29.4	34.8	38.3	36.9	35.8	31.8	31.8	31.9	31.1	28.5	25.3
		MCWB	16.5	18.8	20.9	23.6	26.8	27.8	27.7	27.8	27.5	25.3	21.1	17.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.2	23.0	26.6	28.7	29.5	29.7	29.2	29.2	29.0	28.0	24.6	21.9
		MCDB	25.3	28.9	31.5	36.2	35.7	35.5	32.9	32.8	32.8	31.9	29.4	26.1
	2%	WB	19.8	22.2	25.4	27.8	29.0	29.2	28.5	28.6	28.4	27.5	23.9	20.7
		MCDB	23.8	28.0	30.8	34.1	35.4	35.3	32.2	32.0	32.1	31.1	28.1	25.1
	5%	WB	18.7	21.2	24.5	27.2	28.5	28.7	28.1	28.2	28.1	27.0	23.3	19.8
		MCDB	23.4	27.5	30.2	33.0	34.4	34.5	31.7	31.4	31.6	30.4	27.2	24.6
	10%	WB	17.7	20.1	23.6	26.6	28.0	28.2	27.8	27.9	27.7	26.5	22.7	19.0
		MCDB	22.4	26.6	30.0	32.2	33.3	33.5	31.0	31.0	30.9	29.5	26.3	23.3
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.1	13.6	13.2	10.9	8.4	5.5	5.3	5.8	8.0	10.5	11.5
		MCDBR	13.8	14.7	15.0	15.2	12.8	11.5	7.2	6.8	7.4	8.9	10.9	12.2
		MCWBR	5.6	4.9	4.6	4.6	4.2	3.4	2.7	2.5	2.5	3.0	3.8	4.8
	5% WB	MCDBR	11.2	12.3	13.3	12.8	10.9	9.6	6.1	6.0	6.4	6.9	8.8	10.3
		MCWBR	5.5	5.0	5.1	4.7	4.7	3.5	2.6	2.6	2.7	2.9	3.5	4.5
Clear-Sky Solar Irradiance	taub	0.575	0.542	0.574	0.619	0.680	0.697	0.605	0.585	0.594	0.601	0.633	0.626	
	taud	1.735	1.846	1.807	1.767	1.691	1.676	1.874	1.941	1.901	1.834	1.689	1.680	
	Ebn at Noon	675	746	746	722	676	659	722	738	721	687	622	610	
	Edh at Noon	209	199	216	229	246	248	203	190	194	199	218	215	
All-Sky Solar Radiation	RadAvg	3.71	4.66	5.64	6.23	6.10	5.06	4.30	4.49	4.47	4.43	3.98	3.50	
	RadStd	0.27	0.28	0.24	0.28	0.36	0.39	0.31	0.21	0.24	0.38	0.21	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	+0.45	N/A	N/A	N/A	N/A	N/A	+94

Nomenclature: See separate page